

Mayank Yadav

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PROFESSIONAL SUMMARY

Third-year B.Tech Computer Science & Engineering (Data Science) student at Manipal University Jaipur, with hands-on expertise in Machine Learning, Generative AI, Large Language Models (LLMs), and Agentic AI systems. Shipped five production-grade AI/ML projects spanning mental-health NLP, zero-shot classification, autonomous code-review agents, and cinematic scroll-driven web experiences. Google-certified in Advanced Data Analytics and trained in MLOps pipelines, Transformer fine-tuning, and LangChain agentic workflows. Eager to deepen ML fundamentals and applied research skills through Amazon's AI/ML Summer School.

TECHNICAL SKILLS

Languages & Databases:	Python, SQL, R (basics), HTML, CSS, JavaScript
AI / ML & Modeling:	Machine Learning, Deep Learning, Generative AI, LLMs, NLP, Predictive Modeling, Feature Engineering, Model Optimization, Evaluation Metrics, Reinforcement Learning
Frameworks & Libraries:	PyTorch, LangChain, FastAPI, Flask, Streamlit, NLTK, Hugging Face Transformers, BART, DistilRoBERTa
MLOps & Tools:	Git, GitHub, Docker, Jupyter, Google Colab, VS Code, Tableau (basics)
Core Concepts:	Agentic AI, Prompt Engineering, RAG, Zero-shot Classification, AST Parsing, Scroll-driven UI

EXPERIENCE & VOLUNTEERING

Volunteer — Web & AI Developer | NayePankh Foundation

NGO Volunteer Project · 2025

- Built a Volunteer Information Management System (VIMS) to digitise intake, tracking, and reporting for 200+ volunteers, replacing manual spreadsheet workflows.
- Designed and deployed an AI-related awareness webpage for the foundation, integrating interactive LLM-powered FAQ chatbot capabilities to educate visitors about the NGO's mission.
- Implemented automated email notifications and dashboard analytics using Python (Flask) and SQL, reducing administrative overhead by ~40%.

Software Development Intern | SoftPro India

Internship · 2025

- Engineered a full-stack Guest House Management System covering room allocation, booking workflows, billing, and occupancy reporting.
- Designed RESTful API endpoints with FastAPI and built a responsive front-end UI with dynamic availability calendars.
- Integrated role-based access control (admin / staff / guest) and automated invoice generation, streamlining front-desk operations.

KEY PROJECTS

Health AI Chatbot

Python · NLTK · Transformers · Flask · 2024 · github.com/MAYANK-af/healthai-chatbot

- Engineered a hybrid mental health dialogue engine with a regex-based safety screening layer to instantly flag crisis messages and route users to support lifelines — combining rule-based NLP with deep learning.
- Fine-tuned DistilRoBERTa (Hugging Face Transformers) to classify user emotional sentiment; dynamically adjusted response empathy tokens based on predicted distress level.
- Wrapped conversational flows in a lightweight Flask web app; evaluated model accuracy with F1-score across 7 emotion classes.

FakeNews Radar

Python · FastAPI · BART-Large-MNLI · Docker · JS/Vite · 2024 · github.com/MAYANK-af/fakenews-radar

- Built a zero-shot text classification pipeline using BART-Large-MNLI to evaluate writing style, semantic bias, and linguistic anomalies in news articles — no labelled training data required.

- Developed a dual-layer factual verification engine: semantic Jaccard similarity cross-referencing global fact-checking registries (BOOM Live, PolitiFact) + Google Gemini LLM fallback for live verification.
- Shipped as a Manifest V3 Chrome Extension (highlight-and-scan) with the full backend containerised in Docker; supports real-time inference at the paragraph level.

CodeSense — Autonomous AI Code Reviewer

Python · LangChain · OpenAI API · GitHub API · 2024 · github.com/MAYANK-af/codesense

- Built an agentic LLM pipeline that autonomously reads GitHub PR diffs and flags logic bugs, code smells, style violations, and security vulnerabilities using LangChain tool-use.
- Structured AST parsing pipelines to extract semantic context beyond raw diffs, enabling single-shot generation of comprehensive, structured code reviews.
- Demonstrates core Agentic AI patterns (tool use, multi-step reasoning) directly aligned with Amazon's research in autonomous developer tooling.

ASUS TUF Gaming F15 — Interactive AI Showroom

HTML · CSS · JS · Scroll-Sync · Frame Interpolation · 2026 · huggingface.co/spaces/itachii9090/asus-tuf-showroom

- Developed a cinematic, scroll-driven interactive showroom visualising notebook thermals, GPU architecture, and display specs with hotspot overlays and animated data callouts.
- Engineered custom video-to-frame scroll-sync pipelines for seamless user-controlled render animations, demonstrating applied signal processing for UI interaction.

Porsche 911 GT3 RS — Cinematic Landing Page

HTML · CSS · JS · GSAP · Scrollytelling · 2026 · huggingface.co/spaces/itachii9090/Porsche-911-GT3-RS

- Designed scroll-driven assembly animations powered by frame-interpolated product videos; synthesised custom composite imagery for a narrative scrollytelling experience.

EDUCATION

B.Tech — Computer Science & Engineering (Data Science) · Manipal University Jaipur · 2023 – 2027

Relevant coursework: Machine Learning, Data Structures & Algorithms, Database Management, Operating Systems, Statistics.

Senior Secondary (12th) · St. Paul's Sen. Sec. School, Mathura · 2023

Secondary (10th) · St. Paul's Sen. Sec. School, Mathura · 2021

CERTIFICATIONS

- Google Advanced Data Analytics Professional Certificate (Coursera) — Regression models, ML pipelines, Tableau
- Google Data Analytics Professional Certificate (Coursera) — Data cleaning, SQL, R, visualisation
- Artificial Intelligence A-Z 2024 (Udemy) — Reinforcement Learning, Generative AI, Agentic workflows
- Introduction to Operating Systems (NPTEL) — Multiprocessing, memory allocation, concurrency models

INTERESTS & FOCUS AREAS

Agentic AI Systems · LLMs & Prompt Engineering · NLP & Transformer Architectures · Open Source · Scroll-driven UI Engineering